

Research Design

- **Research design** is a strategy for answering the research questions.
- The way to **collect and analyze research data**.
- **Techniques/tools/theories/algorithms** chosen by a researcher to conduct the research.
- Design aims to ensure valid, and reliable process to **produce meaningful results**.
- In computer science, it varies depending on the nature of study.
- Computer science research design:
 - Mathematical model
 - Simulation
 - Software/application development
 - Hardware
 - Analyze/study
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Research Development/Implementation

- The act of execution or creation or installation or practice of a research design.
- Follows the design exactly.
- In **Computer Science/ICT**, actual implementation of a research design goes through computer programming and other tools.
- In **Social Science or Health Science**, implementation refers to putting a program into practice.
- In **Business Studies**, various types of business plans and initiatives get implemented.

Testing/Experiment/Analysis

- **Testing** is a way of assessing the validity of a research.
- Statistical test, experimental test, software test,
- Software testing is the process of evaluating and verifying that a software product or application does what it's supposed to do.
- Testing checks if implementation works as expected.
- **Experiment & analysis** observes, and measures the research study.
- **Analysis** is a detailed examination of anything with the hope of learning as much as possible.
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Simulation

- Imitation of a situation or a process.
- A simulation is a model that mimics the operation of an existing or proposed system.
- A computer simulation is a program that is run on a computer and that uses step-by-step methods to explore the approximate behavior of a system.
- Example: weather forecasting, climate change, astrophysical simulation, flight simulation software, traffic simulation, virtual reality.....

Result/Outcome/Finding(s)

- This is the core findings of a study derived from the research method.
- This is the final part of a research.
- Draws concluding remarks.
- Shows future path.
- Might be negative which opens a new door.